INCH-POUND

MIL-B-63512 (AR)
AMENDMENT 8
13 JUNE 1989
SUPERSEDING
AMENDMENT 7
20 July 1988

MILITARY SPECIFICATION

BODY ASSEMBLY
FOR
MINE, ANTI-PERSONNEL: BLU 92/B

This amendment forms a part of Military Specification MIL-B-63512(AR), dated 8 March, 1982, and is approved for use within the US Army Armament Munitions and Chemical Command and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 1

2.1, Specifications, Military: Delete "MIL-E-17555" in its entirety.

PAGE 4

* 3.3.1 Transportation vibration. Delete in its entirety and substitute the following:

"3.3.1 Transportation vibration. The Body Assemblies shall be safe to handle and transport. Test Mines shall show no evidence of damage that will effect the functions following transportation vibration Test No. 119, Procedure 2 of MIL-STD-331."

3.3.2, Temperature conditioning of Body Assemblies:

b. Ambient: Delete "for four (4) hours min." and substitute "70°F ± 10°F for four (4) hours min.".

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PAGE 5

* 3.5.2, Torque test of Selector Switch Weldment. Delete: "35 inch pounds" and substitute "24 inch pounds".

* 3.6.1 Quiescent current. Delete in its entirety and substitute the following:

"3.6.1 Quiescent current. The Electronic Assembly Quiescent current shall be less than 240 microamperes at ambient temperature and less than 288 microamperes at cold or hot temperature when the supply voltage is 6.5 ± 0.15 volts and the input/output terminals are loaded as specified in 4.5.7.3."

PAGE 7

* 3.9 Soldering. Delete ",except a specified in Appendix A herein" in its entirety.

* 3.9.1 Solderability. Delete in its entirety and add the following new paragraphs:

"3.9.1 Control of airborne contaminants. Positive air pressure is not required to isolate the soldering area from the rest of the plant to control airborne contaminants.

3.9.2 Soldering irons. Soldering irons shall not attain a temperature greater than 10°F above the control temperature when in an idling or standby condition.

3.9.3 Solderability. Electronic assembly parts that are to be soldered shall comply with the solderability requirements of MIL-S-45743, and MIL-S-46844."

3.10, Add: "f. Detonator and micropiston actuator shorted resistance not greater than 0.15 Ohms".

PAGE 9

* 3.16.2 Electrical workmanship. Delete "Appendix A herein," in its entirety.

PAGE 11

4.3.2.4, Change to read: "...assemblies shall be subjected to transportation vibration in accordance with 4.5.5 and function tested in the mode...".

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TABLE I, FIRST ARTICLE INSPECTION:

Electronic Assembly, Mechanical: Delete "4.4.2.13" and substitute "4.4.2.13.1".

Transistor JANTXV 2N2907: Delete "(Dwg. 9287122)".

Note 2: Delete "... aricle/qualification test" and substitute "... article/qualification test".

PAGE 15

TABLE I, FIRST ARTICLE INSPECTION:

Transistor JNTXV 2N2222A: Delete "(Dwg. 9287122)".

Insert: Delete "(Dwg. 9292629)" and substitute "(Dwg. 9381375)".

Disc, Polyester, Double Coated (Dwg. 9292637): Delete in its entirety.

PAGE 16

TABLE I, FIRST ARTICLE INSPECTION:

Retainer, Coil (Dwg. 9292660): Delete "when used".

PAGE 17

TABLE I, FIRST ARTICLE INSPECTION:

Power Supply Housing Assembly (Dwg. 9292616), Leak Test:

Delete "4.5.3" and substitute "4.5.3.1".

Housing, Upper (Weldment) (Dwg. 9292633):

Delete "(Dwg. 9292633)" and substitute "(Dwg. 9292633-2)".

Housing, Upper (Dwg. 9292635):

Delete "(Dwg. 9292635)" and substitute "(Dwg. 9292635-2)".

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TABLE I, FIRST ARTICLE INSPECTION:

Upper Housing Assembly (Dwg. 9292631), Leak Test:

Delete "4.5.3" and substitute "4.5.3.1".

Electronic and Housing Assembly (Dwg. 9292714):

Delete "Leak test 15 3.5 4.5.3" and substitute "Torque test 15 3.5.2 4.5.3.2".

Body Assemblies (Dwg. 9292713), Functional Tests:

Delete "Functional test ... 4.5.2.3" and substitute "Function test ... "4.3.2.4, 4.5.6".

PAGE 21

4.4.2.2, Delete in its entirety and substitute with new page 21 included.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	OF	DRAWING NUMBER
4.4.2.2	Transistor (JANTX 2N2907A and JANTX 2N2222A)	1	1	MIL-S-19500/255 or 291
				NEXT HIGHER ASSEMBLY: 9292726

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major 1/					
101	Bond wire not aluminum	32	32-0-1	3.2	Visual
102	Compliance with MIL-S-19500/255 or 291		--	3.2	Cert. 2/
103	Solderability (3 leads ea.)	10	10-0-1	3.9.1	MIL-S-45743
Minor 2/					
201	Poor workmanship		0.65%	3.16	Visual

NOTES: 1/ Each transistor type.
2/ Visual evidence of JANTX marking.

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4.4.2.3, Semiconductor Device, Diode, Type JANTX 1N645-1:

Drawing number: Delete "9287168-2" and substitute "MIL-S-19500/240".

Critical 1: Delete in its entirety.

Critical 2: Delete in its entirety.

Add the following Major categories:

"104, Missing or incorrect color band on cathode end of diode - 100% 3.2 Electrical 1/".

"105, Reverse current - 100% 3.2 MIL-STD-750, Method 4016, dc".

PAGE 23

4.4.2.4, Diode, Zener, JANTX IN 4463 and JANTX IN 4471:

Drawing Number: Delete "MIL-S-19500/460" and substitute "MIL-S-19500/406".

Critical 1: Delete in its entirety.

Critical 2: Delete in its entirety.

Critical 3: Delete in its entirety.

Major 103: Delete "MIL-S-19500/460" and substitute "MIL-S-19500/406".

Add the following Major categories:

"104, Missing or incorrect color band on diode (polarity) - 100% 3.2 Electrical 1/".

"105, Reverse current - 100% 3.2 MIL-STD-750, Method 4016, dc".

"106, Zener breakdown voltage V_Z - 100% 3.2 MIL-STD-750, Method 4022".

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4.4.2.6, Capacitor, Fixed, Electrolytic Tantalum Solid (82 uF):

Major 104: Delete "8-0-1" and substitute "8-1-2".

Major 107: Delete "10-1-2" and substitute "10-2-3".

PAGE 26

4.4.2.7, Capacitor, Fixed, Electroytic Tantalum Solid (39 uF):

Major 104: Delete "8-0-1" and substitute "8-1-2".

Major 107: Delete "10-1-2" and substitute "10-2-3".

PAGE 28

4.4.2.9, Switch Assembly:

Next Higher Assembly: Delete "9292725" and substitute "9292726".

PAGE 29

* 4.4.2.10 Wiring, printed, flexible. 101: Delete "MIL-P-50884, 1/" and substitute "MIL-P-50884, Excluding Component Holes, 1/".

PAGE 32

4.4.2.13, Title: Delete "Electronic Assembly (Mechanical)" and substitute "Component Assembly".

Major 101: Delete in its entirety.

Major 110: Delete in its entirety.

After paragraph 4.4.2.13, add paragraph 4.4.2.13.1, Page 32a (included), consisting of Classification of Defects and Tests for Electronic Assembly (Mechanical), drawing number 9292725.

PAGE 34

4.4.2.14 Electronic assembly (Performance characteristics) (Ambient): Add the following Major Category:

"115, TTB Oscillator period, under min. - 100% - 3.6.3 - 4.5.7.2"

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9292725
4.4.2.13.1	Electronic Assembly (Mechanical)				NEXT HIGHER ASSEMBLY 9292714
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Terminals damaged, bent, or canted more than 5 degrees.		0.40%	3.2	Visual/Gage
102	Envelope over max.		0.40%	3.2	Gage
103	Potting height under min.		0.40%	3.2	Gage
104	Components not covered		0.40%	3.2	Visual
105	Radius of lead holder channels incorrect (4 places)		0.40%	3.2	Gage
106	Location of lead holder channels incorrect (4 places)		0.40%	3.2	Gage
107	Conformal coat missing forward of noted area		0.40%	3.2	Visual
Minor					
201	Poor workmanship		0.65%	3.16	Visual

NOTES:

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4.4.2.14 Electric assembly (Performance characteristics) (-40° - -5°F): Add the following Major category:

"115, TTB Oscillator period, under min. - .40% - 3.6.3 - 4.5.7.2"

PAGE 38

4.4.2.14 Electronic assembly (Performance characteristics) (145° + 5°F): Add the following Major category:

"115, TTB Oscillator period, under min. - .40% - 3.6.3 - 4.5.7.2"

PAGE 39

4.4.2.15, Selector Switch Assembly:

Next Higher Assembly: Delete "9292714-2" and substitute "9292714".

PAGE 41

4.4.2.17, Housing Lower, Selector Switch:

Major 112: Delete "Depth of pressure ..." and substitute "Length of pressure ...".

PAGE 42

4.4.2.18: Delete in its entirety and substitute new page 42 included.

PAGE 43

4.4.2.19, Disc, Polyester, Double Coated:

Delete in its entirety.

PAGE 46

4.4.2.22, Post Weld:

Major 104: Delete in its entirety.

* Major 106: Delete "Width of serrations, 3 places" and substitute "Distance between serrations".

* Major 107: Delete in its entirety.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	OF	DRAWING NUMBER
4.4.2.18	Insert, Bushing	1	1	9381375
				NEXT HIGHER ASSEMBLY 9292713

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined	See 4.5.1			
Major	None defined				
Minor					
201	Length		0.65%	3.2	Gage
202	Outside diameter		0.65%	3.2	Gage
203	Diameter of hole		0.65%	3.2	Gage
204	Location of hole		0.65%	3.2	Gage
205	Poor workmanship		0.65%	3.2	Visual

NOTES:

AMSMC Form 1570, 1 Feb 85

Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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4.4.2.24, Cover, Power Supply:

Major 109: Delete in its entirety.

PAGE 51

4.4.2.27, Battery and Lead Assembly:

Major 101: Delete "... lead above case" and substitute "... lead".

Major 106: Delete in its entirety.

Major 107: Delete in its entirety.

Major 108: Delete "*" and substitute "1/".

Add the following Major Category: "109 Battery leaks 2/ 3.2 4.5.2.2".

NOTES: Delete "*Sample size ... " and substitute "1/ Sample size ...".

Add the following NOTE: "2/ Sample size per paragraph 4.5.2.2 herein".

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PAGE 55

4.4.2.31, Retainer, Coil (when used):

Title: Delete "(when used)".

Next Higher Assembly: Delete "9292652" and substitute "9292652/9292713".

PAGE 57

4.4.2.33, Housing Power Supply:

Major 107: Delete in its entirety.

Major 116: Delete "True position of O-ring groove diameter" and substitute "O-ring runout incorrect".

PAGE 58

4.4.2.33 Cont'd, Housing Power Supply:

Minor 201: Delete "Width of tapered slots - 3 for weld posts" and substitute "Width of slots for weld posts (3 places)".

PAGE 59

4.4.2.34, Plate Power Supply:

Major 108: Delete in its entirety.

PAGE 61

4.4.2.35, Power Supply Housing Assembly:

Major 101: Delete in its entirety.

Add the following Major category:

"104, Sensors not covered - 0.40% 3.2 Visual".

PAGE 63

4.4.2.37, Housing Upper (Weldment):

Next Higher Assembly: Delete "9292631" and substitute "9292631-2".

Notes: Delete "insorting" and substitute "inserting".

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4.4.2.38, Housing Upper:

Drawing Number: Delete "9292635" and substitute "9292635-2".

Next Higher Assembly: "9292633" and substitute "9292633-2".

Major 107: After "2 places 5A/4" Add "2/".

Major 116: Delete in its entirety.

Major 117: After "hole 5D/3" Add "2/".

Major 118: After "hole 5D/3" Add "2/".

Major 119: Delete "... of holes in holes inflat ... 4D/4" and substitute "... of holes in flat ... 4D/4 2/".

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4.4.2.38 Cont'd, Housing Upper:

Drawing Number: Delete "9292635" and substitute "9292635-2".

Next Higher Assembly: Delete "9292633" and substitute "9292633-2".

Major 120: Delete "... of holes in holes in flat ... 4 D/4" and substitute "... of holes in flat ... 4 D/4, 2/".

Major 121: Delete "True position of O-ring groove diameter" and substitute "O-ring runout incorrect".

PAGE 66

4.4.2.39, Plate Upper:

Next Higher Assembly: Delete "9292633" and substitute "9292633-2".

Major 105: Delete in its entirety.

Major 106: Delete in its entirety.

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4.4.2.40, Shorting Bar, Ceramic:

Major 101: After "thickness" add "(substrate)".

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Minor 204: Delete in its entirety.

Minor 206: Delete in its entirety.

* Minor 207: Delete in its entirety.

* Minor 208: Delete in its entirety.

PAGE 69

4.4.2.42 Add "Major 101 Conformal coating missing or improper 0.40% 3.2 Visual".

PAGE 70

4.4.2.43, Upper Housing Assembly:

Major 101: Delete in its entirety.

Add the following Major category:

"104 Sensors not covered - 0.40% 3.2 Visual".

PAGE 71

* 4.4.2.44 Cushion. Delete Major 103 in its entirety.

After paragraph 4.4.2.44, add paragraph 4.4.2.44.1, Page 71a (included).

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4.4.2.45, Electronic and Housing Assembly:

Major 101: Delete in its entirety.

* Major 103: Delete "Angularity" and substitute "Location".

Add the following Major categories:

"110 Torque test 1/ 0-1 3.5.2 4.5.3.2".

"111 Selector ball binds or is missing - 0.40% 3.2 Visual/Auditory".

Notes: Delete "(a) Sample per ..." and substitute "1/ Sample per ...".

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.44.1	Shin, Shorting Bar				9381374
					NEXT HIGHER ASSEMBLY 9292713
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined	See 4.5.1			
Major	None defined				
Minor					
201	Length		0.65%	3.2	Gage
202	Length of legs, 2 places		0.65%	3.2	Gage
203	Width		0.65%	3.2	Gage
204	Diameter of hole		0.65%	3.2	Gage
205	Poor workmanship		0.65%	3.2	Visual

NOTES:

AMSMC Form 1570, 1 Feb 85 Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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4.4.2.48, Body Assembly (Prior to Welding the Power Supply Housing Assy and the Electronic/Housing Assy together)

Critical category:

Add the following critical category:

"6 Detonator and micropiston actuator not shorted (prior to soldering) 100% 3.10 Gage".

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Major category:

Major 101: Delete "(Note 2)" and substitute "(Note 2 and Note 30)".

Major 107: Add "(after soldering)".

Major 109: Add "(note 28)".

Add the following major category:

"110 Tape on S&A missing or incorrect (note 31) 0.40% 3.2 Visual".

Major 106: Delete in its entirety.

PAGE 77

4.4.2.49, Body Assembly (Prior to Assembly of Gasket and Epoxy Putty):

Critical: Delete "Not defined" and substitute "None defined".

Add the following Major Category: "110 Shim missing or incorrect 0.40% 3.2 Visual".

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4.4.2.50, Body Assembly (Prior to Assembly Power Supply Cover):

Critical: Delete "Not defined" and substitute "None defined".

PAGE 79

4.4.2.5.1, Body Assembly (Prior to Applying Sealing Compound over Receiver Coil):

Critical 2: Delete "Not defined" and substitute "None defined".

PAGE 80

4.4.2.52, Body Assembly (Final):

Critical 1: Delete "3.7.1" and substitute "3.2".

PAGE 84

* 4.4.6 Launch simulator. Subparagraph a. Delete: "150 + 6" and substitute "145 + 25".

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PAGE 85

4.5.2, Delete in its entirety and substitute the following:

"4.5.2 Battery and lead assembly.

4.5.2.1 Pull test of battery and lead assembly leads. Each lead of each battery and lead assembly shall withstand a non-destructive pull force of 5+2 pounds along the longitudinal axis of the lead without degradation or failure. Additionally one battery at the start of each shift and one battery of every 100 manufactured shall be subjected to a destructive pull force on each of the two leads of 9 pounds minimum along the longitudinal axis of the lead. Evidence of residual lead material shall be present on the battery after the pull test to assure weld integrity.

"4.5.2.2 Leak test of battery and lead assembly. One assembly at the start of each shift and one assembly of every 100 manufactured shall be tested for compliance with the gross and fine leak requirements of MIL-B-48761. Any unit failing to meet the requirement shall reject the lot. (Non-destructive test) (Alternate lead material only)."

PAGE 86

4.5.3.1, Seal Leak Test of the Power Supply Housing Assembly and Upper Housing Assembly: Change "... specified in 3.5 shall ..." to "... specified in 3.5.1 shall ...".

* 4.5.3.2 Torque test of selector switch assy. Delete: "... Dwg 9292679..." and substitute "... Dwg. 9292714".

TABLE II, Body Assembly Electrical Checks Prior to Packaging:

* Electrical check 1: Delete "0.7 Ohms max. 3/" and substitute "0.2 - 1.25 Ohms 3/".

Electrical check 8: Delete "0.8 - 1.4 Ohms" and substitute "0.85 - 1.4 Ohms".

Electrical check 9: Delete "1.0 - 6.0 Ohms" and substitute "2.0 - 6.0 Ohms".

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TABLE II, Body Assembly Electrical Checks Prior to Packaging:

Note 1: Delete "See Dwg. 9292725 ..." and substitute "See Dwg. 9292726 ...".

4.5.6, Performance Tests of Body Assemblies:

Title: Delete "Performance test of body assemblies" and substitute "Performance/Function test of body assemblies".

4.5.6.1, Anti-disturbance (AD) Mode Functioning:

(f): Delete "(see 4.5.6.3.a)" and substitute "(See 4.5.6.4)".

PAGE 88

4.5.6.2, Tripline (TL) Mode Functioning:

(c): Delete "functon" and substitute "function".

(f): Delete "(See 4.5.6.3.a)" and substitute "(See 4.5.6.4)".

4.5.6.3, Self Destruct (SD) Mode"

(d): Delete ("See 4.5.6.3.a)" and substitute "(See 4.5.6.4)".

PAGE 89

4.5.6.5, Recording of performance data:

(b): Delete in its entirety.

(m): Delete "After failure, analysis ..." and substitute "After failure analysis ...".

4.5.7, Electronic assembly tests:

Delete "70° + 5°C" and substitute "70° $\pm$ 10°F".

PAGE 91

4.5.7.8, Low voltage detect test:

Delete "... detonator firing cirucuit when ... 0.15 volts and linearly reduced at a rate ..." and substitute "... detonator firing circuit when ... 0.15 volts and reduced at a rate ...".

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4.5.7.13, Anti-disturbance function mode add the following:

"Alternate procedure. With the electronic assembly under test connected to the test fixture so that its plane is parallel with the plane of the test fixture, verify the anti-disturbance function made by passing a permanent magnet across the top of the AD switch assembly. Proper function will be evidenced by confirming the resulting turn-on of the detonator firing circuit. The magnet may make brief physical contact to the top of the AD switch assembly during this motion. The magnet flux density shall be no less than 20 gauss and no greater than 60 gauss. The magnet holding force shall be 0.7 pounds (minimum). (Non-destructive test)".

PAGE 94

5.1.2: Delete in its entirety.

5.2.1: Delete in its entirety.

PAGE 95

6.7, Data requirements:

Delete "4.4.2.1.4 & 4.4.2.5.4" and substitute "4.4.2.14 & 4.4.2.54".

Delete "4.3, 4.4.2.5.4 & 4.5.6.5" and substitute "4.3, 4.4.2.54 & 4.5.6.5".

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* APPENDIX, Supplemental Soldering Provisions. Delete in its entirety.

The margins of this amendment are marked with an asterisk or vertical lines to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

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